

Gram-Schmidt Orthogonalization Process

$$e_1 = \frac{v_1}{\|v_1\|}$$

Normalize first vector

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

Orthogonalize second vector

$$e_2 = \frac{u_2}{\|u_2\|}$$

Normalize

$$u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j$$

Orthogonalize k-th vector

$$e_k = \frac{u_k}{\|u_k\|}$$

Normalize

Result: $\{e_1, e_2, \dots, e_n\}$ is an orthonormal set